

Basic sound setup

About **\$450** with a borrowed calibrator; **\$700** with our own.

What this buys

- Use our existing computer and brain-recording system.
- Play selected tones below **20 kilohertz**. A higher frequency means a higher-pitched sound.
- A microphone near the ear checks the sound that actually arrives. A sound-level calibrator produces a known test sound to check the microphone.

Computer → **sound interface** → **powered speaker**

Microphone near ear → **sound interface** → **sound recording**

Parts to buy

Blue names are buying links. Exact product codes are kept for ordering; explanations use full words. Prices are in United States dollars, checked October 9, 2026.

Job	Part / seller	Price
Play and record	Behringer UMC204HD – eBay / Sweetwater, new	97.98
Make the sound	One JBL 305P MkII powered speaker – Reverb, new/open box	122.39
Measure sound	Dayton EMM-6 microphone – eBay / Parts Express, new	66.93
Microphone cable	Hosa HMIC-010 microphone cable – Sweetwater	17.99
Audio cables	Two Hosa HSS-005 audio cables , \$14.95 each	29.90
Microphone stand	Gator RI-MICTP-FBM stand – Sweetwater	34.99
Parts	Borrow the sound-level calibrator	370.18
Optional purchase	REED R8090 sound-level calibrator – DigiKey / REED, new	219.00

- **Borrow the calibrator:** \$370.18 parts + \$80 estimated shipping/tax = **\$450.18**.
- **Buy the calibrator:** \$589.18 parts + \$100 estimated shipping/tax = **\$689.18**.
- **Still extra:** any cable or adapter needed to connect timing signals to our brain recorder.

What could be misaligned?

- **Start delay:** storing sound in a computer buffer adds delay. As an example, one buffer of 256 samples at 48,000 samples per second lasts **5.3 milliseconds**; two last 10.7 milliseconds. These are calculations, not measurements of this interface.
- **Sound travel:** the speaker and converters add delay. Air adds about **3 milliseconds per metre** from speaker to ear.
- **Loudness accuracy:** unknown until each tone is checked at the ear.
- **Changing delay between trials:** unknown until we test it. Preload the sound sequence and use wired connections.
- **Drift:** the sound and brain recorders may run at slightly different speeds. Our mismatch is unknown. For scale only, a hypothetical 0.01% speed mismatch would cause **0.36 seconds of error in one hour** without correction.

How we check the timing

- Send a timing reference beside each tone and record that reference with the brain signals.
- Use the microphone to measure the gap between the reference and sound arriving at the ear.
- Recording the microphone only on the sound interface leaves two timelines to align. A sound output is not automatically a compatible digital trigger; check the recorder input.

Sources: [computer buffering](#); [delay estimates](#); [sound travel](#). Sound level must also be checked for each tone at the ear.

Accurate version 1: SONIC paper

\$4,840.36 priced so far; speaker/amplifier quotes and measurement equipment are extra.

What this buys

- Reproduce the paper's sound chain: **sound device** → **speaker amplifier** → **speaker**.
- Add a satellite-based timing reference to help coordinate equipment clocks.
- Reuse our brain recorder. The paper's Connexus brain-recording system is outside this budget.

Parts to buy

One of each. Prices are in United States dollars, checked October 9, 2026. National Instruments makes the sound device; Tucker-Davis Technologies makes the speaker/amplifier. "Quote" means ask the supplier for a price.

Part	Seller	Price
USB-6361 sound device, 782255-01	New / Newark	3,744.42
TimeMachines TM2500C time reference	New / backorder	799.99
MF1-M speaker kit	Supplier quote	Quote
SA1 speaker amplifier	Supplier quote	Quote
ZB1PS powered enclosure*	Supplier quote	Quote
781513-01 power supply*	New / Newark	123.42
780534-01 computer cable*	New / DigiKey	69.75
763830-01 United States power cord*	New / Newark	51.00
781233-02 noise filter*	New / DigiKey	38.25
Hosa BNC-58-103 output lead*	New / Pro Audio Solutions	13.53
Subtotal of priced parts		4,840.36

What is still extra?

- **Sound measurement:** our page-4 kit adds \$1,897.16 with a used microphone amplifier. That brings priced parts to **\$6,737.52**, before the supplier quotes.
- **Connections:** microphone-to-sound-device, network and brain-recorder wiring still need selection. The timing reference is currently backordered.
- **Accessories:** starred parts are our reproduction additions, not exact accessories named by the paper. Confirm included cables, stands and power supplies to avoid buying duplicates.

How precise is it?

- **Pitch range:** the paper tested **1.4–32 kilohertz**. It used about 80 decibels at the ear, but gives no measured sound-level uncertainty.
- **Alignment with brain recording:** the paper gives no measured timing error. The network time reference is specified below one microsecond plus network variation; that is not the accuracy of sound arrival.
- **Drift:** the independent sound-device clock specification corresponds to **180 milliseconds per hour versus ideal time**. This is not the paper's drift with satellite timing connected. Drift against our recorder remains unknown.

Sources: SONIC version 1, Methods 6.1–6.2; sound device specifications; time reference; speaker. A cheaper \$3,498.95 eBay sound-device listing needs exact model/accessory confirmation. The independent-clock figure uses typical conditions at 25 degrees Celsius. The paper used awake sheep; Rice's sedated-sheep adaptation remains unverified. No complete price or upper limit is established.

Accurate version 2: McGinley slides

\$7,551.75 priced so far; speaker quote and measurement equipment are extra.

What this buys

- Reproduce the supplied setup: **computer** → **equipment rack** → **speaker driver** → **speaker**.
- The rack holds a sound board and a separate signal input/output board.
- Reuse our brain recorder instead of buying the slides' Neuropixels system. A compatible desktop computer with a free expansion slot is required.

Parts to buy

One of each. Prices are in United States dollars. Most rack parts are used; several ship from China. Seller claims do not verify operation or calibration.

Part	Seller	Price
PXIe-1082 rack enclosure	Used / eBay	2,650.00
PXIe-6363 signal input/output board	Used / eBay	2,495.36
PXI-4461 sound board, 186900-11L	Used / eBay	699.99
PXIe-8360 rack controller	Used / eBay	493.20
PCIe-8361 computer connection card	Seller-new / eBay	197.00
779500-03 rack cable, 3 metres	Seller-new / eBay	89.85
SHC68-68-EPM cable, 192061C-02	Used / eBay	89.99
BNC-2090A connection panel	Used / no returns	613.77
ED1 driver + enclosure/power supply	Used / eBay	209.06
ES1 speaker + matching cable	Supplier quote	Quote
Hosa BNC-58-103 output lead*	New / Pro Audio Solutions	13.53
Subtotal of priced parts		7,551.75

What is still extra?

- **Sound measurement:** our page-4 kit adds \$1,897.16, making **\$9,448.91 priced so far**. The slides do not identify a microphone or sound-level calibrator.
- **Connections:** microphone/brain-recorder wiring and any missing rack power cord are extra. Confirm the speaker's matching cable is included.
- **Before buying:** confirm the sound board fits the rack and has valid calibration. The driver bundle includes its enclosure and power supply; do not buy another. The starred output cable is our addition.

How precise is it?

- **Pitch range:** the speaker is specified for **4–110 kilohertz**. It misses the paper's lowest 1.4–4 kilohertz tones. The sound board's specified flat output extends to 92.1 kilohertz.
- **Start delay:** the sound board's output filter adds **0.15625 milliseconds** at its highest sample rate. Speaker and air delays are additional. The slides give no measured alignment with brain recording.
- **Drift:** the sound board's internal-clock specification corresponds to **72 milliseconds per hour versus ideal time**, before aging/calibration uncertainty. Clock settings and our recorder determine actual relative drift; that remains unknown.
- **Loudness at the ear:** accuracy is unmeasured. The speaker's uncorrected response variation is ± 11 decibels; that is not calibrated accuracy.

Sources: supplied diagram; supplied parts slide; sound board specifications, pages 19 and 24; speaker specifications; compatible board version. Supplier quote, shipping, tax, software and labor remain extra.

Accurate version 3: my recommendation

\$5,636.41 priced so far; speaker, speaker amplifier and powered enclosure need quotes.

What this buys

- Cover the paper's **1.4–32 kilohertz** range with a compact computer-connected sound device.
- Measure sound with a microphone near the ear. Record repeated timing signals with our brain data to align the recordings.
- This is a proposed build. Its sound level and timing have not been tested.

Parts to buy

United States dollars. One of each except batteries/resistors. Exact product model names are retained for ordering. Main subtotal uses a new microphone amplifier.

Part	Seller	Price
USB-6361 sound device, 781442-01	New / Newark	3,030.42
MF2-M speaker kit	Supplier quote	Quote
SA1 speaker amplifier	Supplier quote	Quote
ZB1PS powered enclosure	Supplier quote	Quote
Earthworks M50 measurement microphone [†]	New / eBay	1,299.00
Sound Devices MM-1 microphone amplifier [†]	New / B&H	945.00
REED R8090 sound-level calibrator [†]	New / DigiKey	219.00
Hosa HMIC-010 microphone cable [†]	New / Sweetwater	17.99
Gator RI-MICTP-FBM microphone stand [†]	New / Sweetwater	34.99
Two Panasonic microphone-amplifier batteries, \$0.59 each [†]	New / Mouser	1.18
Pomona EM4970-36# output lead	New / DigiKey	23.68
Hosa PHX-106F-BULK microphone adapter	New / B&H	13.95
Two Yageo 100,000-ohm resistors, \$0.10 each	New / DigiKey	0.20
763830-01 United States power cord	New / Newark	51.00
Subtotal of priced parts		5,636.41

Ways to reduce cost / what is missing

- **Used microphone amplifier:** the \$325 used MM-1, no returns, brings priced parts to **\$5,016.41**.
- **Shared measurement kit:** marked items cost \$2,517.16 new or \$1,897.16 with the used microphone amplifier. Same kit on pages 2 and 3.
- **Still missing:** supplier quotes and the brain-recorder timing adapter. The microphone adapter needs rewiring before use.

How precise should we expect it to be?

- **Pitch and loudness:** the speaker covers 1–80 kilohertz; the microphone reaches 50 kilohertz. The sound-level calibrator is specified within **0.5 decibels at one kilohertz only**. Accuracy across all tones at the ear remains unmeasured.
- **Drift:** the independent sound-clock specification corresponds to **180 milliseconds per hour versus ideal time**. Use repeated recorded references to estimate and correct the mismatch; remaining error is unmeasured.
- **Timing goal:** **within one millisecond after correction** is our proposed acceptance goal, not achieved performance.

Included: sound-device power supply/computer cable, microphone calibration adapter and speaker stand/cable. Shipping, tax, tariffs, software and labor are extra. No complete price or upper limit is established. Arduino or a field-programmable gate array still needs sound hardware, calibration and timing checks.

Sources: [speaker](#); [microphone](#); [microphone amplifier](#); [sound-level calibrator](#); [sound-device timing](#). Clock figure assumes typical conditions at 25 degrees Celsius. Confirm computer compatibility. Detailed calculations: [supporting comparison](#).